Linear programming

Part VII Selected Topics

Scope

Equivalent models, exact pivots, and cycle prevention establish two-phase simplex. Dual solutions and improving rays certify its three possible outcomes.

Prerequisites

Convex polyhedra, standard form, and duality.

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Chapter route

Learning objectives

- Convert models to standard and slack forms.
- Run simplex from a basic feasible solution.
- Use weak/strong duality and feasibility initialization.
- 29.1 Standard and slack forms
- 29.2 Formulating problems as linear programs
- 29.3 The simplex algorithm
- 29.4 Duality
- 29.5 The initial basic feasible solution

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29.1 Standard and slack forms

Our standard form is $\max\ c^{\intercal}x,\ Ax \leq b,\ x \geq 0$, with $A \in R^{(m \times n)}$. Constraints are non-strict and variables continuous real numbers. Finite binary-input bit complexity assumes rational coefficients. Other texts call equality form standard; check conventions first.

A feasible point satisfies all constraints; an optimum maximizes the objective over feasible points. Infeasible means no feasible point. Unbounded means feasible with arbitrarily large objective, not merely an unbounded feasible region. For example, $\max(-x)$ over $x \geq 0$ has an unbounded region and optimum 0.

Intersections of finitely many linear half-spaces are convex polyhedra:$Ax \leq b$, $Ay \leq b$ imply $A(\lambda x + (1-\lambda)y) \leq b$ for $0 \leq \lambda \leq 1$. Convexity neither guarantees integer solutions nor makes nonlinear products linear constraints.

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Four conversions must recover original variables

Convert minimizing f to maximizing $-f$ and negate the returned objective. Multiply $a^T x \geq b$ by -1. Split equality into opposite inequalities. Replace unrestricted u by $u^+ - u^-$ with both nonnegative. Every u has representation$(\max(u,0), \max(-u,0))$; every such pair recovers an original value by subtraction. Although not one-to-one, the optimum values correspond.

For `min 2u-v; u+v=3; u-2v≥-3; v≥0; u free`, maximize $-2u^+ + 2u^- + v$ subject to

$u^+ - u^- + v \leq 3;\ -u^+ + u^- - v \leq -3;\ -u^+ + u^- + 2v \leq 3$.

Substituting $u = 3 - v$ gives $v \leq 2$ and original objective $6 - 3v$, so $v = 2$, $u = 1$ attains minimum 0, corresponding to maximum 0. Starting with n variables, m constraints, sign splitting uses at most 2n variables and equality splitting at most 2m constraints, adding only constant coefficient bit length.

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Slack form, bases, and basic feasibility

Introduce $s=b-Ax\geq 0$. Let $M=[A\ I_m]$, $q=(x, s)\geq 0$, so $Mq=b$. Slacks are uniquely determined, not freely allocated independent decisions.

Choose m independent columns as basis B and let N be the others. Invertibility gives

$$q_B=M_B^{-1}b-M_B^{-1}M_Nq_N;\ z=v+\rho_N^{T}q_N.$$

Set all nonbasic variables to 0 for a basic solution. It is feasible exactly when $M_B^{-1}b\geq 0$ componentwise. Initial slack basis I is feasible exactly when $b\geq 0$. Basic values need not be positive; zero basic variables cause degeneracy. A fixed basis determines a unique dictionary because the inverse is unique.

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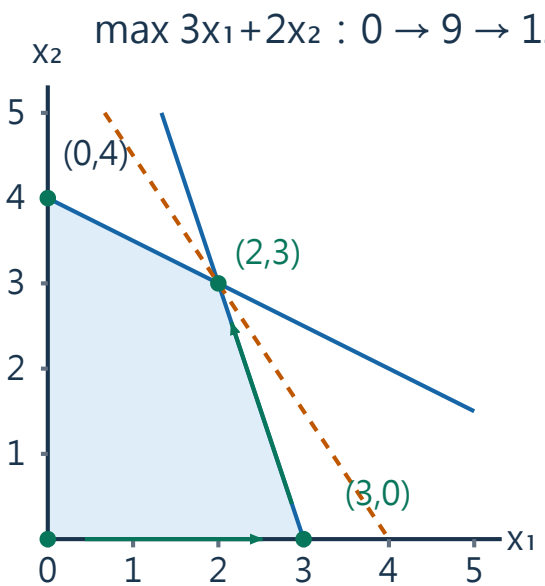
Geometry, level lines, and basis on one example

Use $\max\ 3x_1+2x_2;\ x_1+2x_2\le 8;\ 3x_1+x_2\le 9;\ x\ge 0$. Feasible vertices are(0,0),(3,0),(2,3),(0,4). The vertex path improves objective 0→9→12, ending at(2,3).

Initially $x_3=8-x_1-2x_2;\ x_4=9-3x_1-x_2;\ z=3x_1+2x_2$, with basis{$x_3$, x_4} and slacks(8,9).

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Geometry, level lines, and basis on one example (continued)



Blue: two resource constraints

$x_1+2x_2=8$

$3x_1+x_2=9$

Green: two simplex moves

Orange dashed: $3x_1+2x_2=12$

Intersection(2,3), objective 12

An entire edge can be optimal. For objective x_1+2x_2, the segment(0,4) to(2,3) has value 8. A vertex method returns one optimal basic solution without asserting uniqueness.

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29.2 Formulating problems as linear programs

Specify each variable's meaning, unit, and domain. Map every model-feasible solution to an original plan and every original feasible plan back. Matching objectives while omitting capacity or conservation does not give equivalent models.

For shortest paths, use potentials d_v, fix $d_s=0$, and constrain $d_v - d_u \leq w_{uv}$ per edge. Maximize d_t. Potentials are unrestricted; negative paths can require $d_t<0$.

First assume all vertices are source-reachable, no negative cycle exists, and t is reachable. Summing inequalities along any s-to-t path gives $d_t \leq$ its length. Every feasible potential is therefore a lower bound on the shortest distance. Actual distances $\delta(s, v)$ satisfy all triangle inequalities and attain $\delta(s, t)$. Maximization, not minimization, finds the shortest distance.

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Potential boundaries are not program Infinity values

Restricting to the s-reachable subgraph gives finite true distances when no negative cycle exists. Report unreachable t separately;∞ is not a real LP variable. Keeping an unreachable subgraph with a negative cycle instead makes the whole potential LP infeasible even if the source-to-target distance is ordinary. That assumption cannot be omitted.

If the entire graph has no negative cycle, a zero-edge super-source gives globally feasible potentials. For t unreachable from s, increase every unreachable vertex's potential by arbitrary $K>0$. No reachable-to-unreachable edge exists, so all differences stay feasible. Then d_t is unbounded, corresponding to unreachability, not a negative-length path.

For $s\to a:2$, $a\to t:-1$, $s\to t:4$, constraints are $d_a\le 2$, $d_t\le d_a-1$, $d_t\le 4$. Values(0,2,1) are feasible and the two-edge path bounds $d_t\le 1$, so optimum 1. Erroneously requiring all $d\ge 0$ would exclude correct solutions on other negative-total-path examples.

Flow models use nonnegative variables per original arc

Use arc flow g_{uv} rather than the preceding chapter's antisymmetric net flow. Constrain $0 \le g_{uv} \le c_{uv}$, internal outflow minus inflow 0, and maximize net source outflow. Antiparallel original arcs each retain their own variable and capacity. Do not also impose $g_{uv} = -g_{vu}$.

Every original flow satisfies the model, and capacity/conservation make every model solution a flow. There are m variables and $O(n+m)$ constraints, without n^2 nonedge variables.

Minimum-cost flow additionally fixes source net outflow D and sink net inflow D, and minimizes $\Sigma a_{uv} g_{uv}$. Demand is exact; maximizing flow then reporting cost does not solve the same objective. Finite original capacities bound negative-cost circulation, which need not make the LP unbounded. Prohibiting circulation requires an additional model restriction.

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Complete minimum-cost-flow example

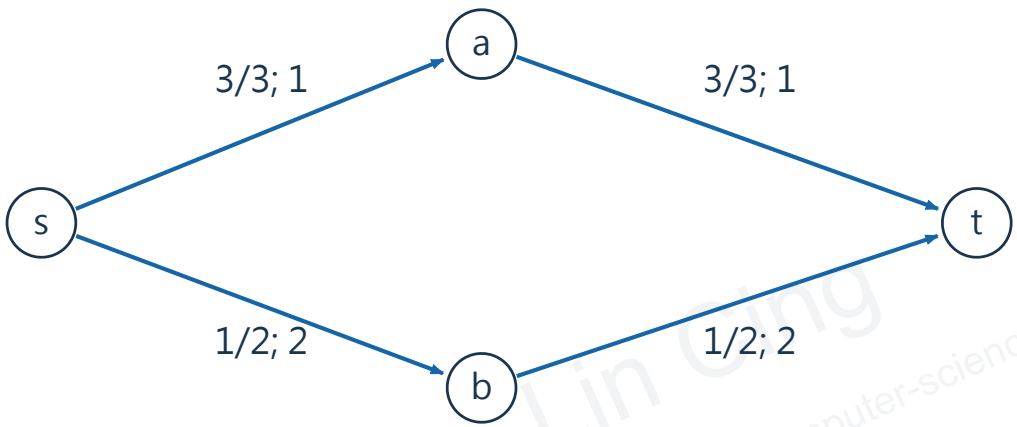
Arcs $s \to a$, $a \to t$ each have capacity 3, unit cost 1. Arcs $s \to b$, $b \to t$ each have capacity 2, unit cost 2. Demand $D=4$. Conservation makes route flows u, v, with $u+v=4, 0 \le u \le 3, 0 \le v \le 2$, and objective min $2u+4v$.

Substitute $v=4-u$ to get cost $16-2u$ and feasible $2 \le u \le 3$. Thus $u=3$, $v=1$ costs 10. This specifies every arc flow, demand, feasible interval, and optimum.

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Complete minimum-cost-flow example (continued)

Demand D=4. Labels: flow / capacity; unit cost.



$u+v=4$

$2 \le u \le 3$

$\text{Cost}=16-2u$

Optimum $u=3$, $v=1$

Upper cost $3\times(1+1)=6$; lower cost $1\times(2+2)=4$; total 10

Using $u+v \le 4$ would incorrectly make zero flow at cost 0 optimal. Omitting $u \le 3$ would incorrectly select $u=4$ at cost 8.

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Multicommodity flow shares capacities

Commodity k has(s_k, t_k, D_k) and per-arc $g^k_{uv} \geq 0$. Each commodity separately conserves flow and meets endpoint net supplies$\pm D_k$. Shared capacity is $\Sigma_k g^k_{uv} \leq c_{uv}$. A feasibility objective may be0; costs can also be included.

Two commodities each uniquely requiring a capacity 1 bottleneck $u \to v$ with demand 3/4 are individually feasible but jointly need $3/2 > 1$. Independent max-flow solutions miss this shared resource.

With k commodities, m arcs, n vertices, there are km variables and $O(kn+m)$ constraints, a polynomial-size LP for rational data. This models splittable flow. One-path-per-commodity or integral-flow requirements create different discrete problems, without the same general LP guarantees.

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29.3 The simplex algorithm

Fix dictionary signs as $x_i=b_i-\Sigma_{(j\in N)}a_{ij}x_j;\ z=v+\Sigma_{(j\in N)}c_jx_j$. Entry a is the coefficient being subtracted, opposite the displayed signed coefficient. All ratios and pivots use this contract.

From feasible $b\geq 0$, choose $c_e>0$ to enter, set $x_e=\theta\geq 0$, and keep other nonbasics 0. Basic $x_i=b_i-a_{ie}\theta$. Rows $a_{ie}\leq 0$ impose no upper bound; positive a_{ie} requires $\theta\leq b_i/a_{ie}$. Choose the minimum nonnegative ratio with a positive denominator,**including zero**.

If no denominator is positive, direction $q_e=1$, $q_i=-a_{ie}\geq 0$, others 0 gives feasible ray $x(\theta)=x(0)+\theta q$ with objective $v+c_e\theta\to\infty$. Do not write $x_e=\infty$; infinity is not a feasible real value.

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Pivoting retains the leaving variable's coefficient

For leaving l and $a_{le} \neq 0$, solve its row exactly:

$x_e = b_l/a_{le} - x_l/a_{le} - \Sigma_{(j \in N \setminus \{e\})}(a_{lj}/a_{le})x_j$.

Substitute into every other row and objective, then exchange B, N. Variable x_l becomes nonbasic, not absent. Omitting $-x_l/a_{le}$ changes the feasible set.

New entries are $a_{el} = 1/a_{le}$, $b_e = b_l/a_{le}$, $a_{ej} = a_{lj}/a_{le}$. For $i \neq l$, $b'_i = b_i - a_{ie}b_e$, $a'_{il} = -a_{ie}/a_{le}$, $a'_{ij} = a_{ij} - a_{ie}a_{lj}/a_{le}$. The objective becomes $v' = v + c_e b_e$, $c'_l = -c_e/a_{le}$, $c'_j = c_j - c_e a_{lj}/a_{le}$.

This invertible substitution preserves every point satisfying the equations and its objective, not merely the current basic point. With nonnegativity, it defines the same LP.

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General feasibility proof for the ratio test

The new entering value $b'_e=\theta=b_l/a_{le}\geq 0$. Other $b'_i=b_i-a_{ie}\theta$ stay nonnegative:for positive a_{ie}, the minimum ratio bounds θ; for nonpositive a_{ie}, the new value is at least $b_i\geq 0$. With new nonbasics 0, the new basic solution is feasible.

Objective improvement is $c_e\theta\geq 0$, strict for $\theta>0$. At $\theta=0$, the point and objective may stay unchanged while the basis changes. This proves both why negative-denominator rows do not restrict the step and why zero ratios cannot be excluded.

```
while some reduced cost c[e]>0:
    choose entering e
    candidates={i in B : a[i, e]>0}
    if candidates empty: return feasible point + improving ray
    l=minimum-ratio row using a specified anti-cycling rule
    pivot(l, e)
return current basic solution
```

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Two pivots on the same two-dimensional example

Initially B={3,4}. Enter x_1:ratios 8/1=8,9/3=3 make x_4 leave with θ=3:

$$x_1=3-x_2/3-x_4/3;\ x_3=5-5x_2/3+x_4/3;\ z=9+x_2-x_4.$$

Enter x_2:ratios 3/(1/3)=9,5/(5/3)=3 make x_3 leave:

$$x_2=3-3x_3/5+x_4/5;\ x_1=2+x_3/5-2x_4/5;\ z=12-3x_3/5-4x_4/5.$$

Nonbasic $x_3=x_4=0$ gives(2,3), tight in both constraints. Every reduced cost is nonpositive, so every feasible point has objective$\leq$12. No further numerical search of the polygon is needed.

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Zero steps are legal pivots

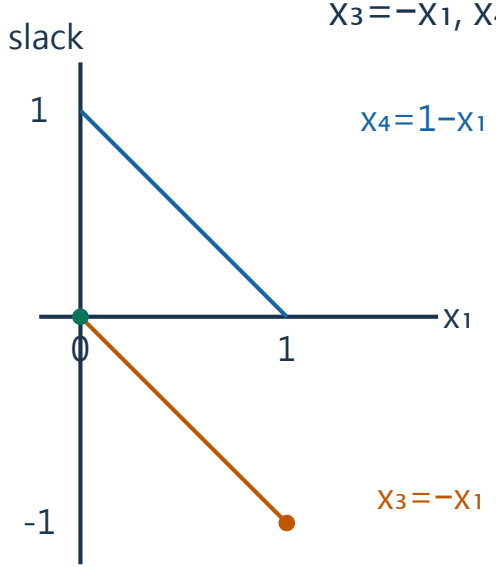
Use $\max\ z=x_1;\ x_3=-x_1;\ x_4=1-x_1$, all variables nonnegative. Initially B={3,4}, N={1}, basic values(0,1).

Basic row	Positive denominator	Ratio	At θ=0	Incorrect θ=1
x_3	1	0	0	−1, infeasible
x_4	1	1	1	0

Let x_3 leave. New $x_1=-x_3$, $x_4=1+x_3$, $z=-x_3$ has the same point(0,0,1), objective 0, but now nonpositive reduced costs prove optimality. Ignoring zero ratios immediately violates feasibility.

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Zero steps are legal pivots (continued)



$x_3 = -x_1$, $x_4 = 1 - x_1$: do not skip a zero ratio

$x_1 = 0$: $x_3 = 0$, $x_4 = 1$, feasible

Ratios: $0/1 = 0$ and $1/1 = 1$

Legal step $\theta = 0$; x_3 leaves

Wrong $\theta = 1$: $x_3 = -1$, infeasible

Only $x_1 = 0$ is feasible

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Finitely many bases alone do not prevent cycling

Matrix M=[A I] has m rows, and a basis selects m independent columns. There are at most C(n+m, m) possibilities. A fixed basis determines one dictionary, so an infinite run repeats a basis. Degenerate steps may preserve objective, so mere nondecrease does not forbid repetition.

Use exact lexicographic perturbation for a self-contained anti-cycling proof. From an initial feasible dictionary $q_B=\beta-Dq_N$, replace its RHS by $\beta_i+\varepsilon^i$ for $i=1,\ldots, m$. Here ε is a positive infinitesimal,**not an arbitrary floating decimal**. Nonnegative constants and a positive first ε coefficient make the initial dictionary strictly feasible in the ordered field.

Represent comparisons by coefficient vectors:constant first, then $\varepsilon,\varepsilon^2,\ldots$; the first nonzero coefficient determines sign. Continue minimum-ratio tests with positive denominators, using later coefficients to resolve equal real ratios.

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Why the lexicographic rule terminates

Let the initial equation matrix be $M_0=[I\ D]$, with RHS $\beta+\Sigma e_i\varepsilon^i$. For any invertible basis K, its basic values are

$$M_0,\ K^{-1}\beta+\Sigma(M_0,\ K^{-1}e_i)\varepsilon^i.$$

No inverse row is entirely zero, so no basic value is the identically-zero polynomial. Feasibility therefore makes every basic value strictly positive. Two positive-denominator ratios cannot coincide identically, or two inverse rows would be proportional, contradicting invertibility. Every legal θ is strictly positive, and $c_e>0$ makes the lexicographic objective strictly increase.

A fixed basis has one dictionary/objective, so it cannot repeat. Finite bases imply termination. Taking constant coefficients returns an original-LP feasible basic point. Reduced costs do not depend on RHS; nonpositive costs establish its optimality. An improving ray likewise has ε-independent direction and certifies original unboundedness from the constant feasible point. No guessed sufficiently-small floating ε is required.

A reproducible lexicographic tie

Use $x_3=2-x_1;\ x_4=2-x_1-x_2;\ \max\ x_1+x_2$, with RHS $2+\varepsilon, 2+\varepsilon^2$. Entering x_1 compares $2+\varepsilon$ against $2+\varepsilon^2$; the latter is smaller, so x_4 leaves.

New rows are $x_1=2+\varepsilon^2-x_2-x_4;\ x_3=\varepsilon-\varepsilon^2+x_2+x_4;\ z=2+\varepsilon^2-x_4$. No reduced cost is positive. The constant solution is $x_1=2$, other variables 0, objective 2. Although x_3 has zero constant, $\varepsilon-\varepsilon^2$ is strictly positive and separates degeneracy symbolically.

Another common anti-cycling choice is Bland's rule:smallest-index positive reduced-cost entering variable, and smallest-index leaving variable among minimum-ratio ties. Both parts matter. The complete termination proof here uses lexicographic perturbation and does not claim arbitrary choices terminate.

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Per-pivot cost is not total simplex cost

A dense ordinary dictionary updates O(mn) coefficients per pivot. This does not prove polynomial total time; basis count C(n+m, m) can be exponential. Lexicographic RHS adds m+1 coefficients per row and a polynomial update/comparison factor, but retains only a finite/exponential general worst-case pivot guarantee.

Rational exact pivots use fractions. Basis solutions are ratios of subdeterminants. After clearing original denominators, a determinant has at most m! terms, giving bit length bounded polynomially by dimension and input bit length. A single basis thus needs no unrestricted representation growth, but pivot count need not be polynomial.

Floating implementations need separate feasibility/reduced-cost tolerances, near-zero ratio handling, and residual recomputation. Clamping negative b to 0 can change the problem. Ellipsoid and interior-point methods are different algorithms; their polynomial theory does not follow from these pivot counts.

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29.4 Duality

Primal $\max\ c^{\intercal}x,\ Ax \leq b,\ x \geq 0$ has dual $\min\ b^{\intercal}y,\ A^{\intercal}y \geq c,\ y \geq 0$. Its m resource rows give m prices and n primal variables give n price inequalities. Transpose A; merely exchanging max/min is insufficient.

For feasible x, y, nonnegativity and both constraint systems give

$$c^{\intercal}x \leq x^{\intercal}A^{\intercal}y = y^{\intercal}Ax \leq y^{\intercal}b.$$

This is weak duality. A dual feasible point bounds every primal objective above. Equality certifies optimality on both sides. Equal objective numbers without checking feasibility are not a certificate.

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Constructing a dual from the terminal basis

Let $M=[A\ I]$, $d=(c,0)$, and terminal basis matrix M_B be invertible. Define row $y^T=d_B^T M_B^{-1}$. Substituting basic variables gives

$z=y^T b+\Sigma_{(j\in N)}(d_j-y^T M_j)q_j$.

For every column, reduced cost $\rho_j=d_j-y^T M_j$, and basic costs are 0. Terminal $\rho\le 0$ gives $-y_i\le 0$ from slack columns e_i, hence $y\ge 0$. Decision columns give $c_j-y^T A_j\le 0$, hence $A^T y\ge c$.

The current feasible basic objective is $y^T b$, attaining weak-duality equality. A finite optimal primal therefore has a same-value dual. Read y_i as the negative reduced cost of original slack i, or0 if that slack is basic. This follows from the full dictionary identity, not coefficient matching at only one point.

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The same example's dual and complementary slackness

The two-dimensional example has dual $\min\ 8y_1+9y_2;\ y_1+3y_2\ge 3;\ 2y_1+y_2\ge 2;\ y\ge 0$. Terminal $z=12-3x_3/5-4x_4/5$ gives y=(3/5,4/5). Both dual rows are tight, and 24/5+36/5=12 matches x=(2,3).

The general gap decomposes exactly as

$b^Ty-c^Tx=y^T(b-Ax)+x^T(A^Ty-c)$.

For feasible sides, every summand is nonnegative. Gap 0 is therefore equivalent to all products y_i(b_i−A_ix)=0 and x_j((A^Ty)_j−c_j)=0. This proves necessity and sufficiency of complementary slackness. Positive primal slack forces its price 0; positive primal variable forces its dual row tight.

For $\max\ 2x_1+x_2;\ x_1+x_2\le 4;\ x_1\le 2;\ x\ge 0$, dual y=(1,1) gives upper bound 6. Point(2,2) has value 6 and all complementary products 0. Point(1,3) has value 5, with the sole positive product $y_2(2-x_1)=1$.

The same example's dual and complementary slackness (continued)

$$\text{gap} = y^T(b - Ax) + x^T(A^Ty - c) \geq 0$$

$x=(2,2)$

$y=(1,1)$, dual value 6

primal 6

$y_2(2-x_1) = 0$

Other three products are 0

$x'=(1,3)$

$y=(1,1)$, dual value 6

primal 5

$y_2(2-x_1) = 1$

Other three products are 0

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Unboundedness and infeasibility are not interchangeable

An unbounded primal has no feasible dual, or one fixed dual value would bound all primal values. A downward-unbounded dual similarly makes primal infeasible. The converse need not hold:an infeasible primal need not have an unbounded dual, because both can be infeasible.

Primal $\texttt{max}-x;\ 0x\leq -1;\ x\geq 0$ is infeasible; dual $\texttt{min}-y;\ 0y\geq -1;\ y\geq 0$ is feasible and downward-unbounded. Changing primal objective to max x gives dual constraint $0\geq 1$, making both sides infeasible.

For other sign conventions, convert to standard first. A primal equality has a free dual multiplier; an unrestricted primal variable gives a dual equality. These rules follow by combining split nonnegative variables or opposite inequalities.

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29.5 The initial basic feasible solution

Negative entries in initial b prevent starting a ratio loop that assumes feasibility. For $Ax \leq b$, $x \geq 0$, add $a \geq 0$ and solve auxiliary `max -a; Ax -a1≤b`. It is always feasible:take x=0, a=max(0,−min_i b_i). Its objective is at most 0 and cannot be upward-unbounded.

Original feasibility allows the same x with a=0, attaining auxiliary optimum 0. Conversely, auxiliary optimum 0 gives a=0 and an original feasible x. A strictly negative optimum proves infeasibility. A currently positive artificial variable alone is insufficient; first establish auxiliary optimality.

When m=0, use the origin directly. A positive c_j gives an unbounded coordinate ray; otherwise the origin is optimal. No minimum over an empty row set is needed.

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A special pivot initializes auxiliary feasibility

Write $s_i=b_i-A_ix+a$ and choose r with minimum $b_r<0$. Let a enter, s_r leave. The pivot coefficient is -1; this is **not** the improving ratio step from an already-feasible basis.

Solving gives $a=-b_r+A_rx+s_r$; other rows become $s_i=b_i-b_r+(A_r-A_i)x+s_r$. Set new nonbasic $x=s_r=0$. Then $a=-b_r>0$ and $s_i=b_i-b_r\geq 0$, establishing a feasible basis. Only now use simplex and lexicographic anti-cycling.

Start symbolic RHS perturbation at this feasible dictionary and take constant parts afterward. Reduced costs ignore RHS, so the result is the original auxiliary optimum and feasible point, rather than a false infeasibility judgment from an ε objective.

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Zero artificial value still requires basis cleanup

At auxiliary optimum 0, delete a's column if it is nonbasic. If a is basic with value 0, its row is `a=−Σa_ajx_j`. A nonzero coefficient permits that x_j to enter and a to leave. This zero-step algebraic pivot leaves the point unchanged; its coefficient may have either sign and it is not an improving ratio step.

If every coefficient is 0, the equation a=0 is redundant. Delete its row and a. General equality or multi-artificial implementations need this branch. Full independent slack rows can sometimes exclude it by rank, but never justify division by 0.

Finally substitute every currently basic variable appearing in the **original objective** to recover a nonbasic-only objective. Changing a title or retaining auxiliary−a does not restore the original problem. This completes Phase I; Phase II begins with that feasible basis and restored objective.

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Complete Phase I and original-objective example

Original $\min\ z=2x_1+x_2;\ x_1+x_2\geq 5;\ x\geq 0$ uses surplus $x_3\geq 0$ with $x_1+x_2-x_3=5$. Add artificial $a=5-x_1-x_2+x_3$, minimizing $w=a$ in Phase I, equivalent to maximizing $-a$.

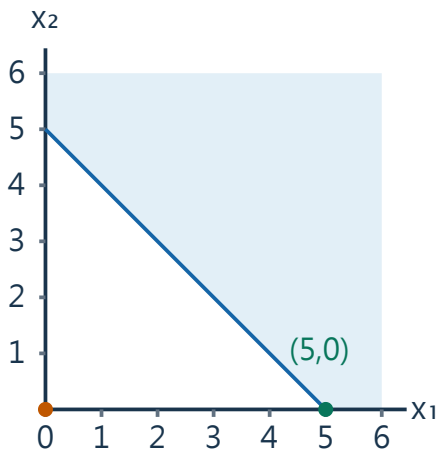
State	Basic row	At nonbasics 0	Objective
Initial	$a=5-x_1-x_2+x_3$	$a=5$	$w=5$
x_1 enters, a leaves	$x_1=5-x_2+x_3-a$	$x_1=5$	$w=a$, min 0
Delete a; substitute original objective	$x_1=5-x_2+x_3$	$x_1=5$	$z=10-x_2+2x_3$
x_2 enters, x_1 leaves	$x_2=5-x_1+x_3$	$x_2=5$	$z=5+x_1+x_3$

Nonnegative final minimization coefficients prove optimum 5. Original origin(0,0) is infeasible; adding $a=5$ makes it feasible only for the auxiliary problem. This one-equation example uses one artificial column; the shared-a construction above covers arbitrary $Ax\leq b$ systems.

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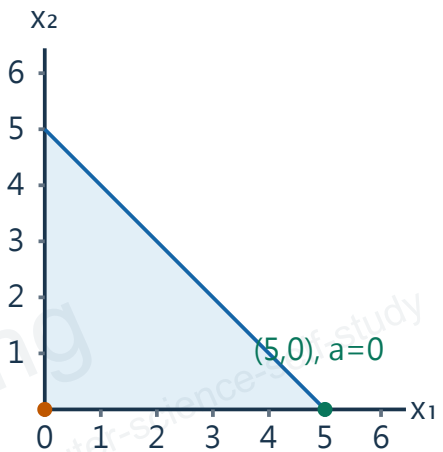
Complete Phase I and original-objective example (continued)

Original: $x_1 + x_2 \geq 5$



Origin infeasible

Auxiliary slice $x_3 = 0$: $a = 5 - x_1 - x_2$



Origin with $a = 5$ is feasible

Artificial variables change only the auxiliary problem. Setting $a = 0$ restores original constraints.

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Infeasibility needs a completed certificate

Original $x \le 1$, $x \ge 2$, $x \ge 0$ becomes $x - a \le 1$, $-x - a \le -2$. Summing gives $-2a \le -1$, so $a \ge 1/2$. Values $x = 3/2$, $a = 1/2$ satisfy both tightly, proving auxiliary $\max -a = -1/2 < 0$ and original infeasibility.

The complete dictionary starts $s_1 = 1 - x + a$, $s_2 = -2 + x + a$. Special a-entry, s_2-exit gives $a = 2 - x + s_2$, $s_1 = 3 - 2x + s_2$, $w = -2 + x - s_2$. Enter x:ratios 2,3/2 make s_1 leave, producing

$$x = 3/2 - s_1/2 + s_2/2;\ a = 1/2 + s_1/2 + s_2/2;\ w = -1/2 - s_1/2 - s_2/2.$$

Both reduced costs are nonpositive, so this is truly optimal. A positive artificial amount alone would not suffice. The original inequalities also sum to $0 \le -1$, an independent contradiction certificate.

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Three outcomes and the proof dependencies

Phase I starts from a feasible auxiliary basis. Lexicographic simplex terminates finitely, and auxiliary objective is bounded above. A negative optimum means infeasible; a zero optimum permits artificial cleanup and objective restoration.

Phase II also starts feasible and terminates under the lexicographic rule. No positive ratio denominator produces a feasible point plus improving ray, certifying unboundedness. Otherwise nonpositive terminal reduced costs produce a feasible basic point and same-value dual, certifying a finite attained optimum.

Thus every LP in this standard form is exactly one of infeasible, feasible with unbounded objective, or feasible with a finite attained optimum. Finite-dimensional structure and non-strict constraints matter. Problem $\max\ x;\ 0 \le x < 1$ has unattained supremum 1 and lies outside this trichotomy's assumptions.

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References

Bibliography

Chapter source

Cormen, Leiserson, Rivest, Stein. *Introduction to Algorithms*,3rd ed., MIT Press,2009, Chapter 29.

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References and attribution

Thomas H. Cormen, Charles E. Leiserson, Ronald L. Rivest, Clifford Stein. Introduction to Algorithms, 3rd Edition.

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